

Deep Learning-Based Analysis of Ancient Hebrew Sources

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Introduction

Authorship attribution and stylistometric analysis examine whether a text contains measurable writing-style patterns. In this project, we focus on Ancient Hebrew sources, especially the Five Books of the Torah. The goal is not to make an absolute authorship claim, but to build a computational research pipeline that converts textual units into stylistic signals, compares them, and detects similarities, transitions, and anomalous regions. The project builds on the Deep Impostors methodology, where separator models are trained on external Hebrew texts called impostors. These trained models are then applied to the target corpus in order to examine how the Torah is represented from a stylistic point of view. In Phase B, the theoretical framework from Phase A was implemented as two complementary pipelines: a CNN/FastText-based Deep Impostors pipeline and a Hebrew BERT Sen2Pro-style probabilistic pipeline.

Mathematical Core

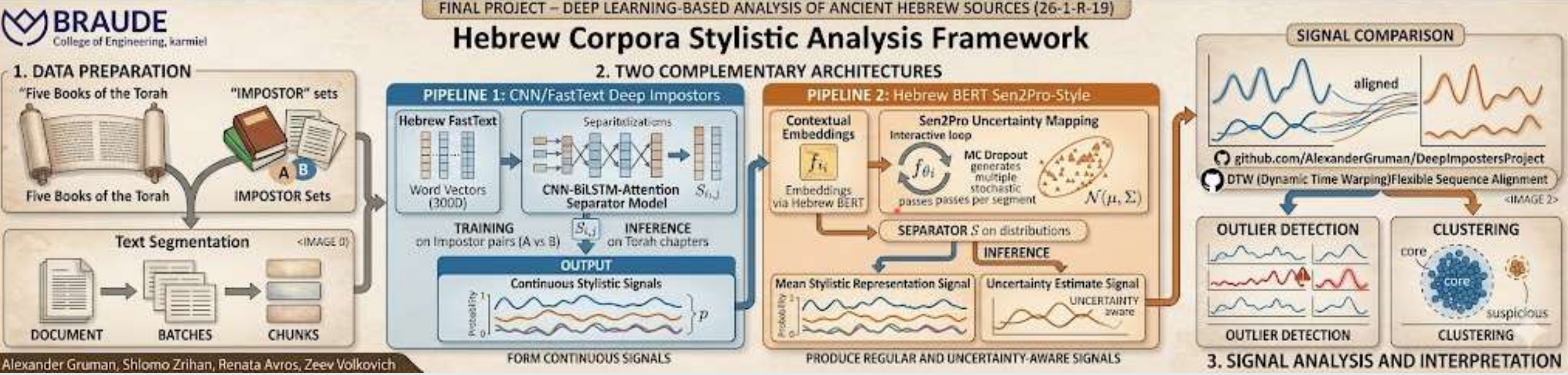
Let a document D be divided into fixed-size batches $:D = (b_1, b_2, \dots, b_n)$
For an impostor pair (A, B) , a separator model $h_{A,B}$ is trained. Each target batch receives a probability score: $p_i = h_{A,B}(b_i)$, $p_i \in [0,1]$. A style signal is created by averaging every r consecutive batch scores: $s_j = (\frac{1}{r}) \sum p_i$. Therefore, each document becomes a numerical signal: $S(D) = (s_1, s_2, \dots, s_m)$. Distances between textual units are computed using FastDTW: $M_{u,v} = FastDTW(S(D_u), S(D_v))$.

Sen2Pro-Style Representation

In the **Sen2Pro-style** pipeline, each text segment is passed through Hebrew BERT several times with MC Dropout enabled.
For a segment x: $z_t = f\theta drop(x)$, $s.t: t = 1, \dots, T$ The representation is estimated by its mean vector: $\mu = (\frac{1}{T}) \sum z_t$ and its uncertainty is estimated from the sample variance/covariance: $\Sigma = (\frac{1}{T-1}) \sum (z_t - \mu)(z_t - \mu)^T$. High variance indicates an unstable representation, while low variance indicates that the model represents the segment consistently.

Methodology

The system processes Hebrew corpora through a multi-stage research pipeline: preprocessing, text segmentation into fixed-size windows, vector representation, separator model training on impostor authors, and inference on Torah chapters to extract stylistic signals.
Two complementary architectures were implemented:
1. CNN/FastText Deep Impostors Pipeline
Representations: 300-dimensional Hebrew FastText word vectors.
Architecture: CNN-BiLSTM-Attention separator model trained on pairs of external impostors.
Output: Probability scores per text unit that form continuous stylistic signals.
2. Hebrew BERT Sen2Pro-Style Pipeline
Representations: Contextual embeddings via Hebrew BERT.
Uncertainty Mapping: Monte Carlo (MC) Dropout generates multiple stochastic passes per segment.
Output: Extracts both a mean stylistic representation and an uncertainty estimate, producing regular and uncertainty-aware signals.



Challenges Review

- **Adapting to Hebrew:** Hebrew's rich morphology, connected words, ancient structures, and source differences required careful preprocessing.
- **Running Time:** Large Hebrew FastText files and repeated experiments required caching, intermediate saving, and resume mechanisms.
- **Result Stability:** Results had to be tested across different impostor pairs, data orders, and parameters, not only one successful run.
- **Sen2Pro Cost:** Probabilistic representation required several samples per segment and made the Sen2Pro version more computationally expensive.

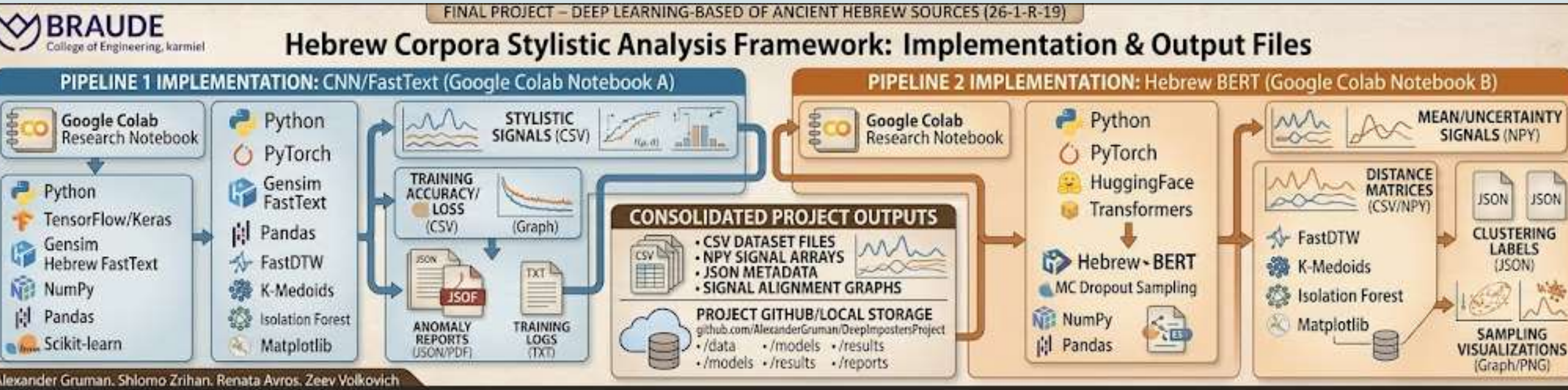
Validation & Reproducibility

Before full experiments, a single-pair sanity check was used to verify that the complete pipeline works correctly. The validation process included:

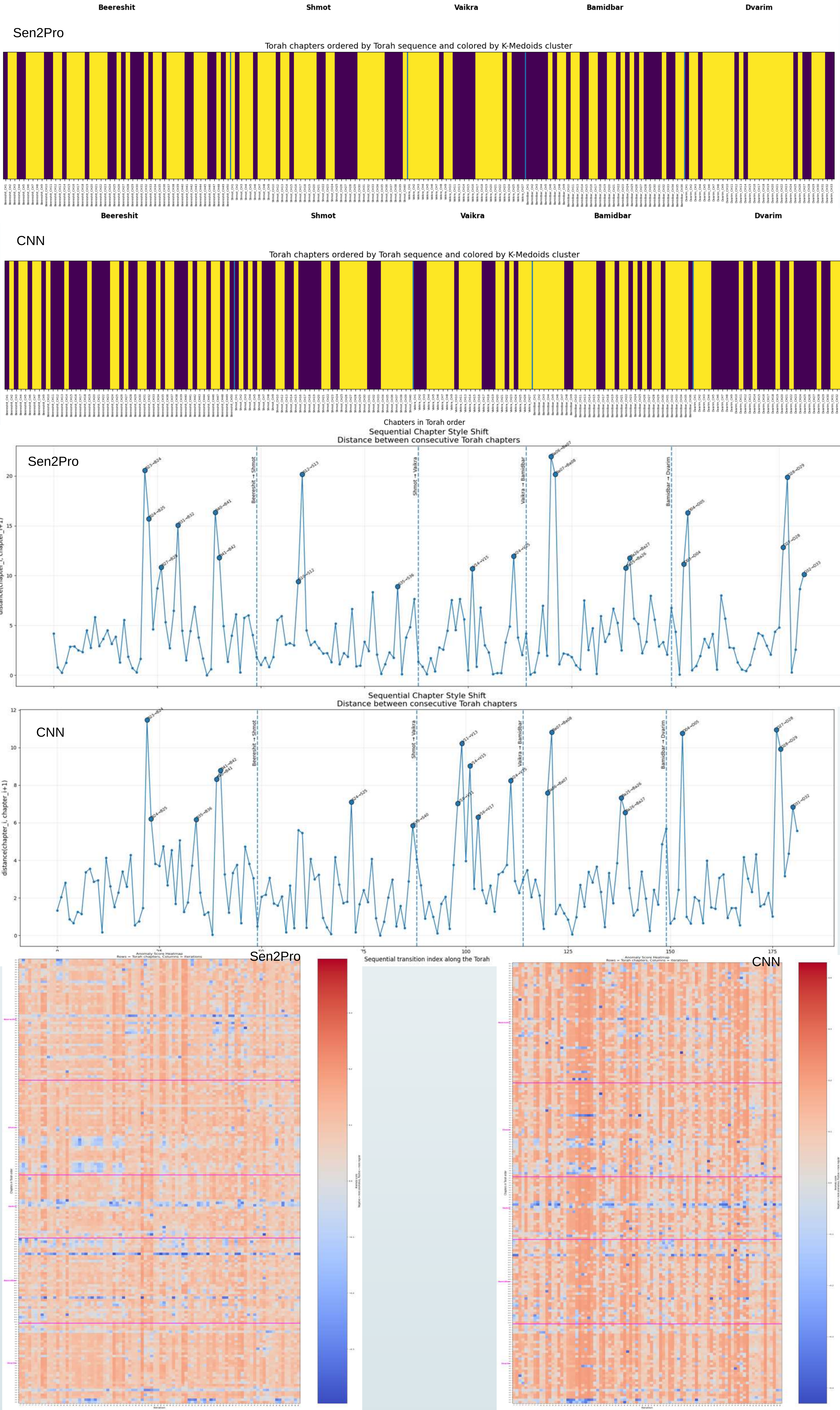
- Checking that input paths and corpus folders are loaded correctly
- Verifying Hebrew text cleaning and segmentation
- Checking tensor shapes and segment counts
- Verifying balanced labels between impostor authors
- Confirming that output files are created successfully
- Saving progress.json to resume experiments after Colab disconnections
- Repeating experiments across different impostor pairs to examine result stability

Implementation & Technologies

The system was implemented as two Google Colab research notebooks.
CNN Pipeline: Python, TensorFlow/Keras, Gensim, Hebrew FastText, NumPy, Pandas, Scikit-learn, FastDTW, K-Medoids, Isolation Forest, and Matplotlib.
Sen2Pro-Style Pipeline: Python, PyTorch, HuggingFace Transformers, Hebrew BERT, MC Dropout sampling, NumPy, Pandas, Scikit-learn, FastDTW, K-Medoids, Isolation Forest, and Matplotlib. The outputs are saved as graphs, CSV files, NPY files, JSON metadata, style signals, distance matrices, clustering labels, anomaly reports, and training accuracy/loss plots.



Results



Key Results:

- **K-Medoids Clustering:** Torah chapters were divided into two clusters. The division did not fully match book borders, suggesting internal stylistic changes inside the books.
- **Sequential Chapter Style Shift:** Both methods detected strong stylistic transitions between neighboring chapters. Similar peaks appeared around Genesis 23–24, Numbers 6–7, Deuteronomy 4–5, and Deuteronomy 28–29.
- **Anomaly Heatmap:** Repeated horizontal patterns appeared across many iterations, meaning that some chapters consistently received higher anomaly scores.
- **CNN vs. Sen2Pro:** CNN produced sharper local changes, while Sen2Pro-style representations produced smoother and more stable patterns. The agreement between the two methods strengthens confidence in the detected structures.

Conclusion

This project implemented a working research pipeline for stylistometric analysis of Ancient Hebrew texts. The system combines Deep Impostors, style-signal extraction, FastDTW-based comparison, K-Medoids clustering, and Isolation Forest anomaly detection. Two different implementations were compared: a CNN/FastText pipeline and a Hebrew BERT Sen2Pro-style probabilistic pipeline. Both methods produced similar chapter-level patterns, supporting the conclusion that the pipeline captures consistent stylistic structures rather than only random model noise. Future work can extend the system by adding more Hebrew corpora, testing additional distance metrics, using different window sizes, adding fuller Sen2Pro data uncertainty, and integrating uncertainty directly into clustering and anomaly detection.

Acknowledgments

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